

ALU

Function

Make the operations, ADD, SUB, AND, OR, RLC B, RRC B, SETC, CLRC, NOT B, NEG B, INC B, DEC B, bypass B, update all flags (Z, N, C, V).

Inputs

- A[7:0]
- B[7:0]
- carry
- op[3:0]

Outputs

- out[7:0]
- Z
- N
- C
- V

CCR

Function

Register file 4 addresses each address is 1 bit to store all the flags where [0] = Z, [1] = N, [2] = C, [3] = V.

Inputs

- rst
- en
- Z
- N
- C
- V
- sel[1:0]
- flags_in_en
- flags_in[3:0]

Outputs

- flag
- flags_out[3:0]

Status_Reg

Function

Save CCR in case of interrupt.

Inputs

- flags_in_en
- flags_in[3:0]

Outputs

- flags_out[3:0]

ALU_OP

- ADD = 0
- SUB = 1
- AND = 2
- OR = 3
- RLC B = 4
- RRC B = 5
- SETC = 6
- CLRC = 7
- NOT B = 8
- NEG B = 9
- INC B = 10
- DEC B = 11
- bypass B = 12